

London Petro Academy Ltd

Mini MBA: Renewable Energy Markets, Auctions & Project Finance

Understanding power markets, renewable policies, tender design and bankable project structures through a week-long competitive bidding simulation.

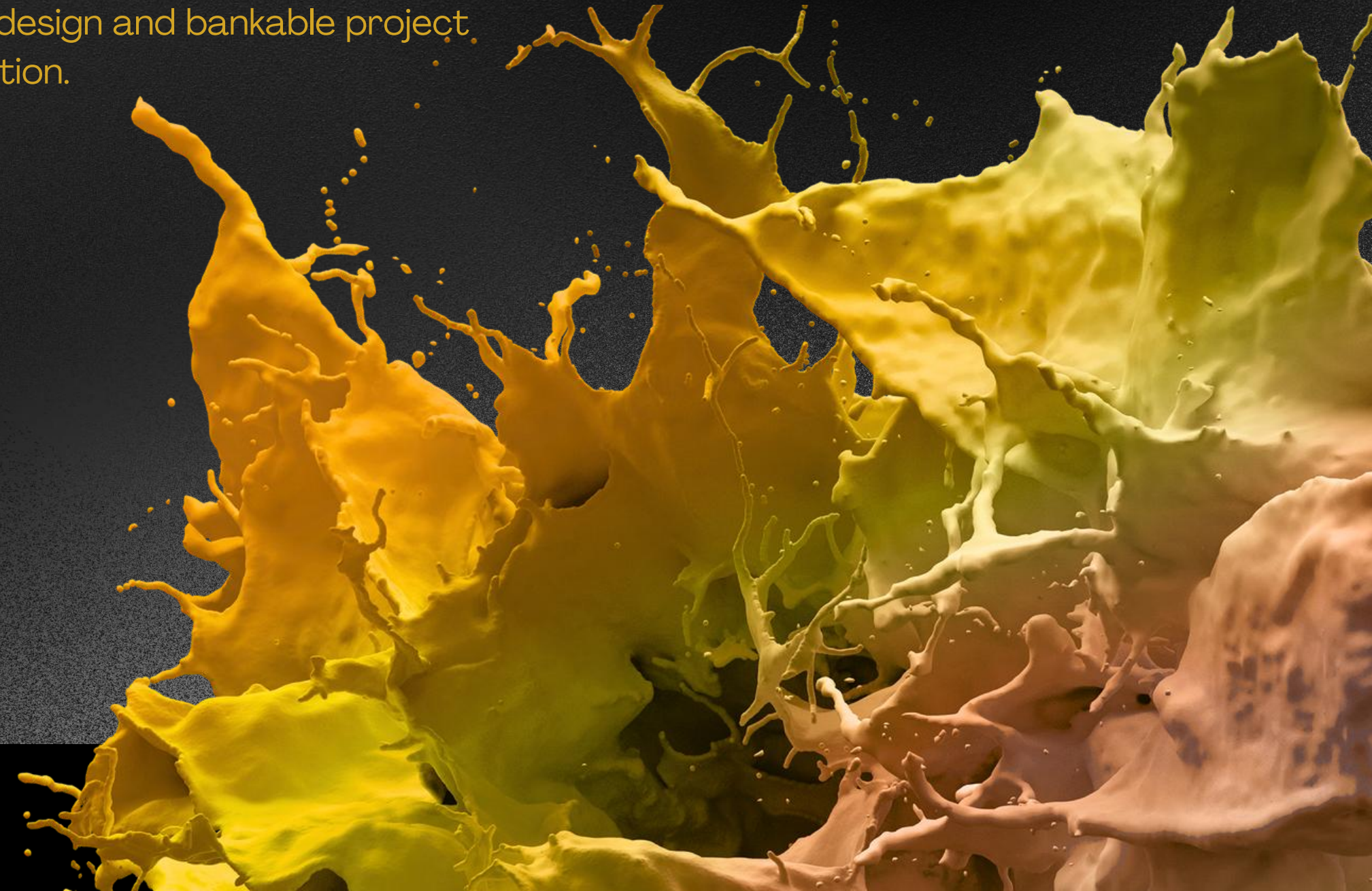


LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future

2026/2027

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Course Overview

This five-day Mini MBA: Renewable Energy Markets, Auctions & Project Finance provides a comprehensive, practice-oriented understanding of how renewable power projects compete, win tenders and achieve bankable financing in today's evolving electricity markets. It begins with global energy dynamics, electricity market design and pricing, grid integration and storage, and the policy instruments that shape renewable investment (feed-in tariffs, certificates, tax incentives and government auctions). Participants then explore technology trends across wind, solar, CSP, bio-power and geothermal, assessing resource uncertainty, cost trajectories and competitiveness. Building on this, the course introduces project finance concepts, loan term sheet elements and key contractual tools (PPAs, EPC and O&M agreements, insurance and hedging) used to mitigate risk and secure debt. Throughout the week, delegates work in groups on a realistic government tender, selecting projects and consortia, modelling economics and finance, raising equity, and presenting bids, culminating in a competitive tender evaluation and "winner" announcement.

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COURSE OBJECTIVES

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By the end of this course, participants will:

- Understand how electricity markets and pricing work, how renewables affect wholesale prices, capacity markets and grid operations, and how policy instruments create or limit opportunities.
- Gain a solid grasp of major renewable generation technologies, resource assessment, energy uncertainty and the factors driving capacity factors, production and cost trends.
- Learn core project finance concepts for renewables, including cash-flow waterfalls, DSCR and other coverage ratios, loan term sheet structures, and the roles of banks, DFIs and ECAs.
- Recognise the contractual and risk-mitigation building blocks of bankable projects, including PPAs, EPC and O&M contracts, insurance and financial hedging instruments.
- Apply these concepts in a live tender simulation: analysing project options, structuring finance, forming consortia, raising equity from different sponsor types, and preparing competitive bids.

The group tender work is a central learning device, ensuring that concepts are immediately applied and debated, rather than remaining abstract.

Course audience

This course is aimed at professionals who want an integrated view of the renewable power sector and how markets, technology, finance and risk come together in competitive project bidding. It is suitable for corporate strategy and business development teams, project managers, sales managers and engineers from oil and gas, utilities and IPPs, institutional investors and private equity, public and private banks, manufacturing, engineering and construction firms, and law practices.

Group Work – Renewable Energy Tender:

Based on real world tenders, delegates will receive on the first day documents of a fictitious government asking international developers and investors to participate in a competitive bidding process. Each day we will reserve the second afternoon session to develop the proposals till each group is able to finalize their bid and present it on the last day. Participants will have to make decisions from a selection of projects, EPC and O&M offers and find an optimal project finance solutions from commercial bank loans, Export Credit Agencies and Multilateral Banks and raise equity from third party investors. Participants are encouraged to apply their new knowledge, share their own experience and have heated discussion within their group to enhance the practical learning experience! In the final day, the groups will present and discuss the various proposals and a tender winner will be declared receiving a surprise prize!



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Content Calendar

Day 1

Markets

Electricity Markets & Pricing

Grid Integration & Storage

Renewable Policy

GROUP WORK:
Tender Introduction

Day 2

Technologies

Wind & Solar

Energy Uncertainty

CSP, Bio-power &
Geothermal

GROUP WORK:
Project Economics

Day 3

Project Finance

Debt Finance

Loan Term Sheet Elements

GROUP WORK: Financial Analysis

GROUP WORK:
Financial Analysis

Day 4

Risk Mitigation

PPA Agreements

EPC & O&M contracts

Insurance & Hedges

GROUP WORK:
Tender Consortia

Day 5

Project valuation

Equity Sponsors

Project Transactions

GROUP WORK: Fund
raising

GROUP WORK:
Tender Evaluation

Electricity Market Drivers & Renewables

We will begin this day in setting the context of the evolution of the global energy market and highlight the current energy dynamics and mega drivers. Over the day we will develop an understanding of how electricity markets work, how prices are set, how energy policy influences renewables and the challenges of grid integration and storage.

Electricity Markets & Pricing

- Understanding Levelized Cost of Electricity and short and long-term marginal cost
- Capacity and energy market prices and long-term Power Purchase Agreements
- Wholesale power market generation and difference to retail markets
- Competition within a wholesale electricity market and the Merit Order effect
- Capacity markets and the adjustment of energy-only market structures
- New reality of power systems – large percentage of renewables
- Impact of intermittent renewables on wholesale markets and other generation assets

Grid Integration & Storage

- What are the barriers to renewable grid integration and the possible solutions?
- What are the learnings from national energy transitions in Germany, China and US?
- Costs of renewable integration and how they can be shared
- What are the answers to growing curtailment threats in saturated energy markets?
- Facilitating Renewable Integration by Demand Response, Smart Grid and energy storage?
- Technological breakthroughs in energy storage as a potential game changer for the energy industry
- Virtual power plants and distributed power generation
- Electricity storage and distribution strategies – can electrical vehicles offer a solution?

CASE STUDIES from Germany, Denmark, UK, Spain and the US demonstrating the ground-breaking electricity market transition with large percentage of renewable generation.



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Renewable Energy Policy Instruments

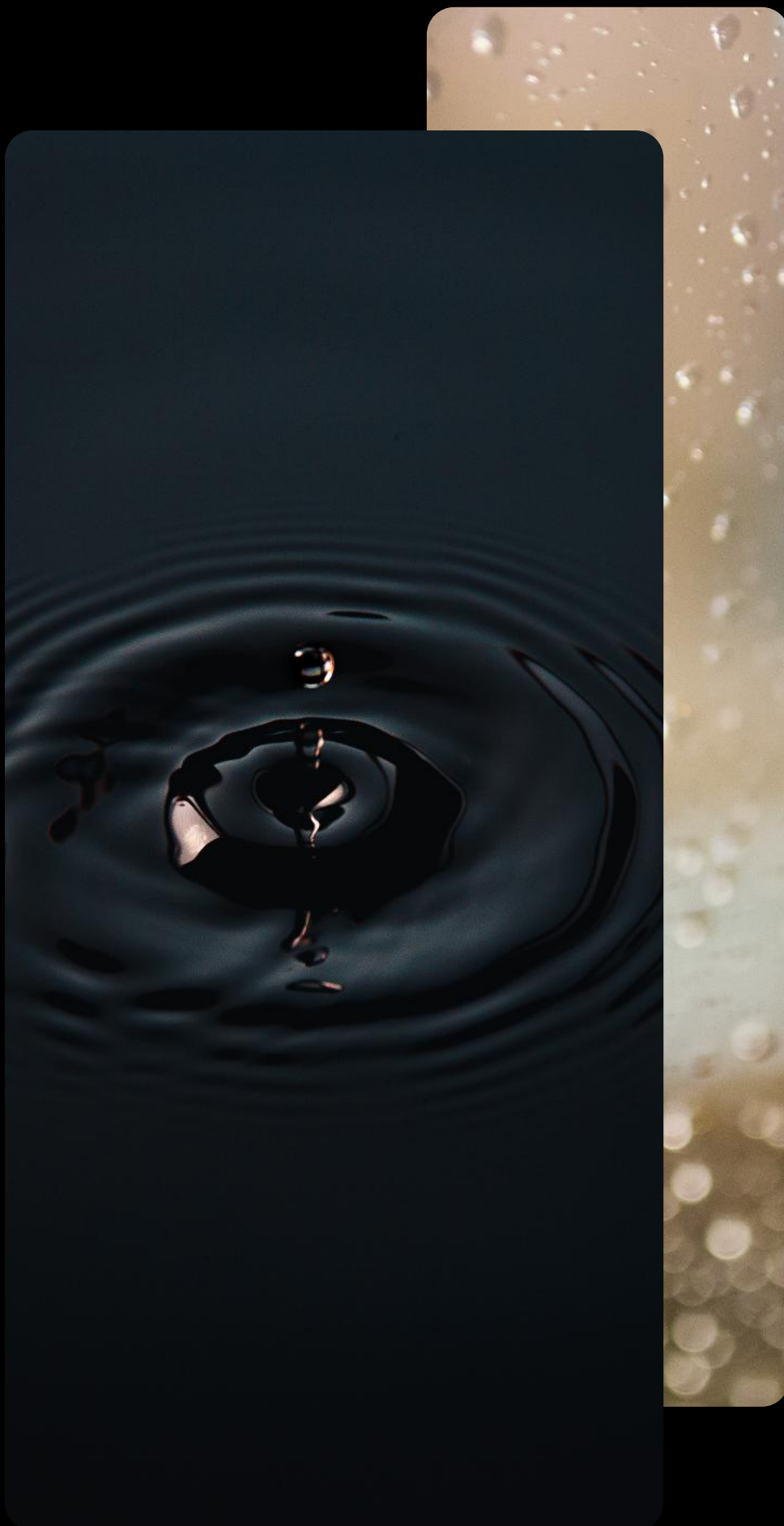
- Understanding feed-in tariffs, certificates, tax incentives and government auctions
- Impact on returns and debt financing of the various policy instruments
- Policy support compared: fossil fuels and renewable energy technologies
- Which policies and regulations have been successful and how well they translate across regions?
- Changing policy needs of emerging and maturing markets
- Slow down of new policy adoption and retrospective changes to national regulatory frameworks
- Emerging trends in renewable policy and their impacts on investors' risk perception
- Policy incentives as main drivers of renewable energy development and investment in the past
- Experience and differences with government auctions and tenders in Europe, Africa and America
- Local content or consortia requirements as a means for job creation and knowledge transfer

CASE STUDIES: How government auctions are driving electricity prices and capacity growth to lowest levels in the world – a model to adopt elsewhere?

Group Work: Tender Introduction

- Tender documents will be distributed, objectives and content explained
- Delegates will form small groups mixing professional experience and cultural background
- Delegates to structure tasks ahead and work plan for the week

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Renewable Power Technologies

This day will develop a solid understanding of the underlying concepts and current status of the different renewable power generating technologies. We will further examine costs trends and key challenges for future developments and their economic impact on competitiveness of each technology.

Wind & Solar Photovoltaic

- Rotor blades as a key item for wind turbines defining power curve and energy production
- Drive trains and how their reliability impacts cost of energy and service costs
- Larger rotor diameters, increasing tower heights – is the sky the limit?
- Are the teething problems of offshore technologies overcome or why offshore at all?
- Variables of capacity factors and availability and their differences
- From silicon, ingots, wafer, cells to modules and solar PV power plants – limits of growth?
- Silicon, thin film, multijunction, organic and concentrated photovoltaics – where is the future?
- Balance of system including inverters, transformers, cabling and structural components
- Watt-peak, cell efficiency, performance ratio and standard test conditions – Watt means what?
- One-axis and two axis tracking system – increased energy output or headache

Energy Uncertainty & Resource Assessment

Extracting wind and solar energy – the Betz' law and Shockley–Queisser limit

- From global irradiation to direct, diffuse and horizontal irradiation
- Satellite and terrestrial data available for wind and solar resource assessment
- Measurement campaigns in order to acquire bankable annual energy production estimates
- Uncertainties of power curves or it wasn't like this in the brochure
- Understanding the resource assessment report – standard deviation, variance and p50/p90
- Experience of predictability and variability of wind and solar resources assessments over time
- Bankability impact of using one-year or ten year data and additional measurement investments
- How reducing uncertainties of annual energy output drives debt capacity and financial returns
- Sources of uncertainties and variability – expertise or layman understanding of the variables

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Bio-Power; Geothermal & Concentrated Solar Power

Co-firing versus dedicated biomass power plants for combined heat and power

- From combustion steam cycle to Stirling engine, ORC, BICGT/BIGCC
- Biogas via anaerobic digestion for small scale and off-grid installations
- Pelletisation, torrefaction, pyrolysis and gasification – where are they commercially?
- Bankable feedstock supply for biomass power plants – quality, quantity, price and on time
- Flash & Dry steam plants, Binary Kalina cycle(ORC) plants and their applications
- Challenges in drilling and exploration – Limits of scalability for geothermal power plants
- Parabolic troughs, towers, linear Fresnel and parabolic dish – which concept where?
- Where are the economies of scale within concentrating solar technologies?
- Use of molten salts as thermal storage for shifting production to cover evening peaks

Group Work: Project Economics

- Understanding resource audits and annual energy production estimates for various technologies
- Using online data tools to assess and interpret different resource potentials
- Modelling different renewable technology projects using SAM and RETScreen
- Understanding key drivers of Annual energy production, economics and capacity factors

Project Finance

In our third day of the seminar we will look at how the various pieces of market and technology fit together and how they can achieve a project finance solution to deliver returns to investors. Participants will have the entire afternoon to work through exercises using sample financial models in order to calculate key sponsor and lenders ratios and see how they react to changing input variables.

Debt Finance

- Development of project and bond finance within the renewable sector
- Project financing structures and the complex commercial relationships and legal issues
- What means a bankable project and how to understand and meet lender's ratios
- Risk phases: Development, Construction, Start-up and Operations & Maintenance
- Trigger Points: Concession signing, Financial Close, Commercial Operation Date, End of Loan
- Multilateral banks and export credit agencies – essential door openers for emerging markets

CASE STUDIES: Various innovative project finance and bond finance solutions will be used to discuss the approach and rational for the stakeholders

Loan Term Sheet Elements

- Capital injections into the project, pro rata basis and timing of drawdowns
- Additional lenders security using Grace Period, Debt Service and O&M Reserve Accounts
- Cash Sweeps, Cash Traps and Dividend Blockers and how they damage Equity returns
- Where the various elements relate to within the Cash Flow Waterfall
- Covenants, Securities and the concept of Sources and Uses
- Interest Rate, Banking Fees, Margins and Commissions applied to a Project Finance Loan
- Conditions Precedents – Important steps towards Financial Close

Direct Agreements

CASE STUDIES: Analysing content and structures of different loan term sheets of real world projects

Group Work: Financial Analysis

- Continue using SAM, RETScreen and introduction of Excel based financial models
- Understanding the Cash Flow Waterfall
- Calculation of CFADS, DSCR, LLR, PLCR and how to interpret minimum and average ratios
- How the Cash Flow, the DSCR and loan tenor drives overall debt capacity and equity returns
- Understanding Equity NPV and IRR drivers and calculate LCOE
- Modelling the tender project with various sensitivity analysis and combining loans

This course is available In-House and can be tailored to your specific requirements.



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Touchpoints

Question about this course?

Call us on +44 (0) 1582 516247 and one of our representatives will assist you with your enquiry.

3 Easy ways to register

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Email: info@londonpetroacademy.co.uk

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